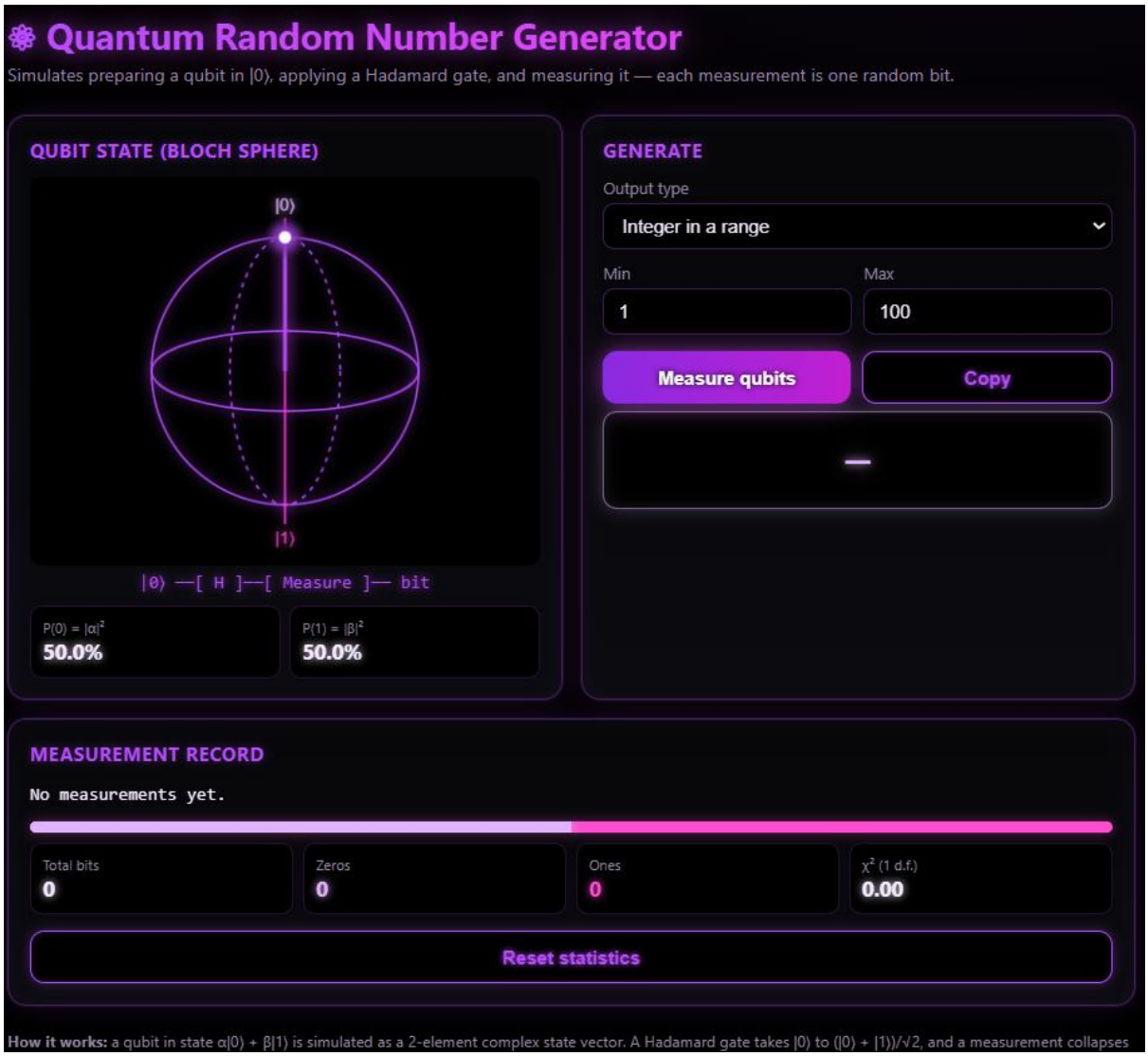
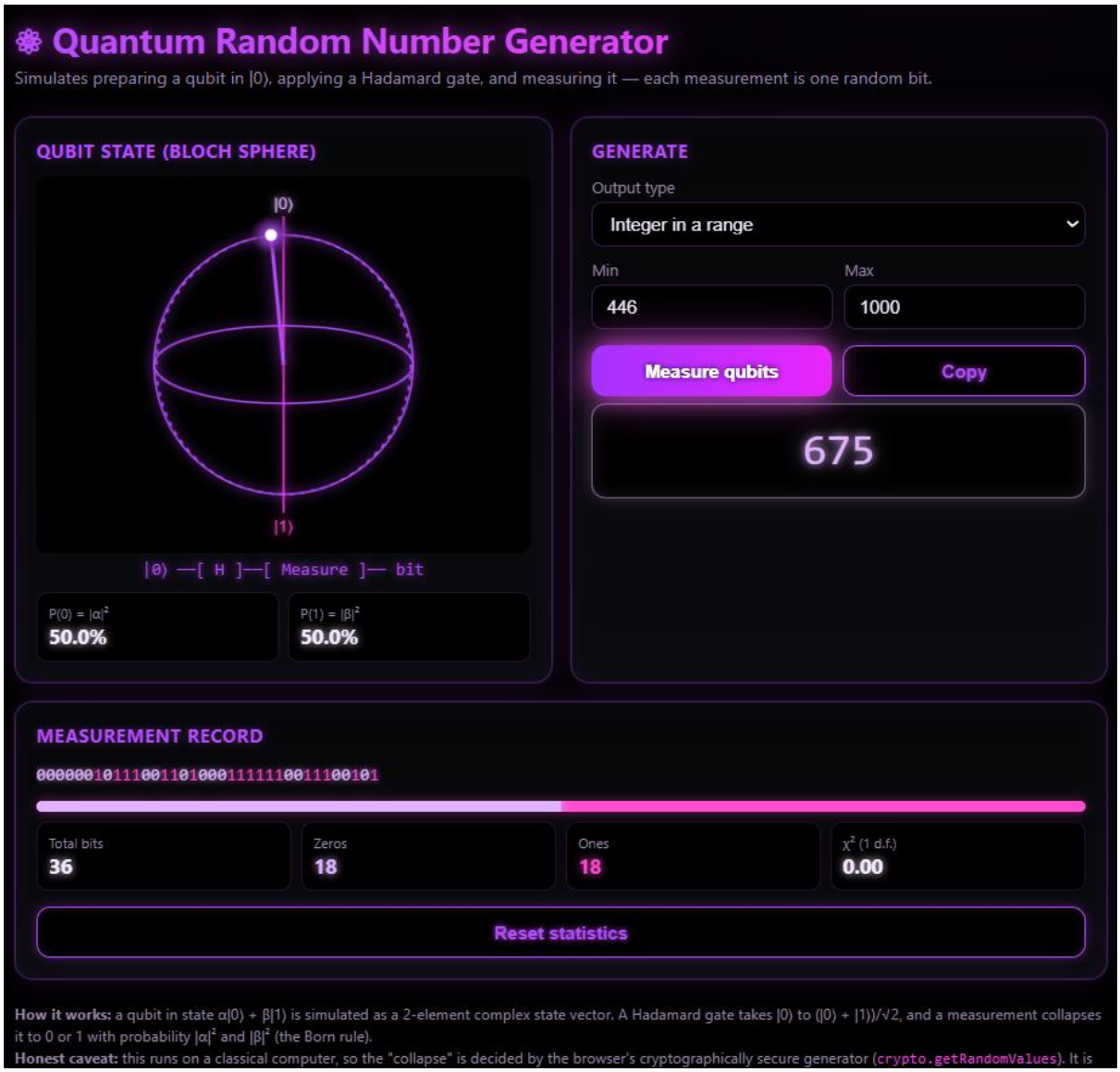
Quantum Random Number Generator

UI:-

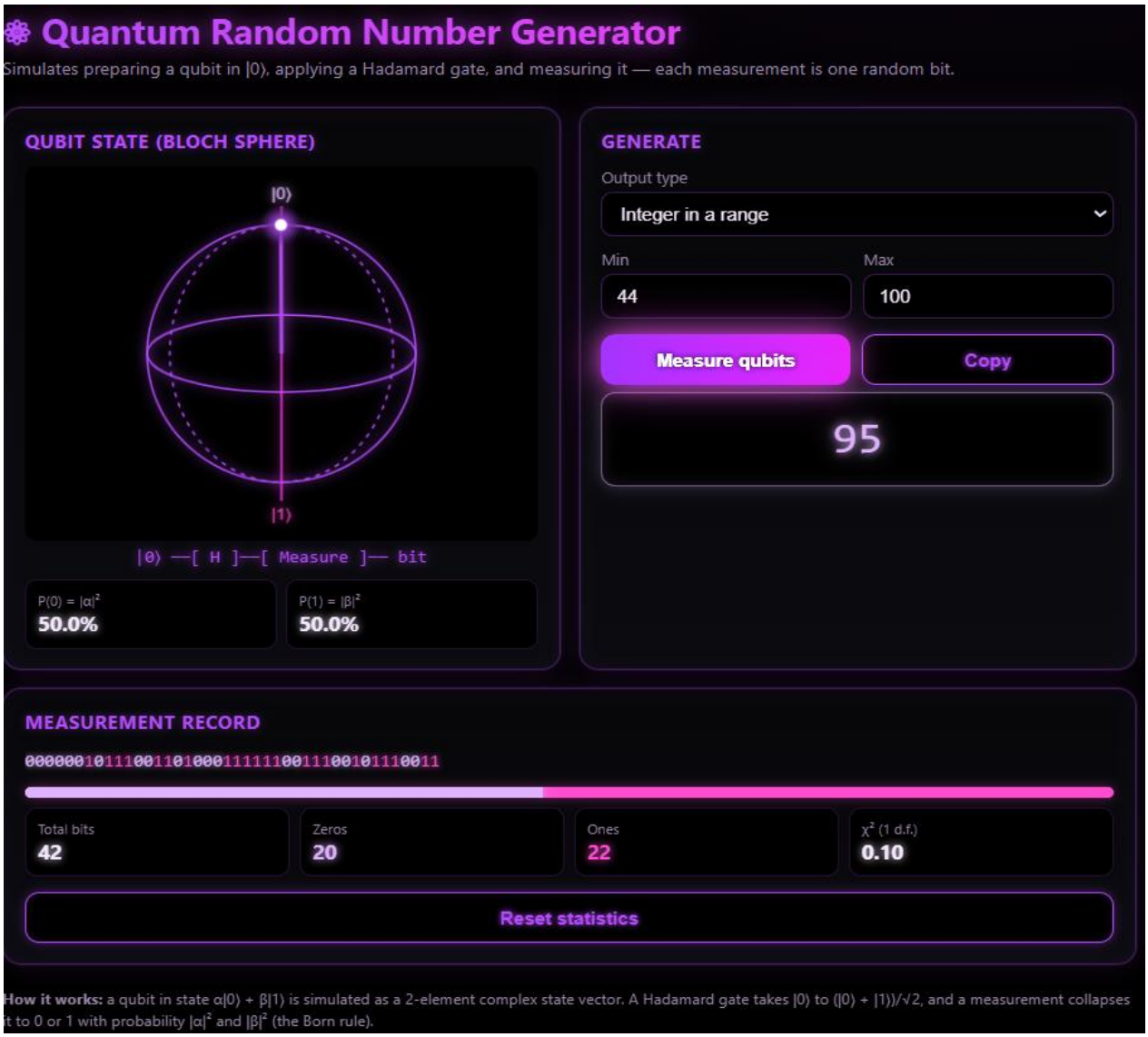


Results:-

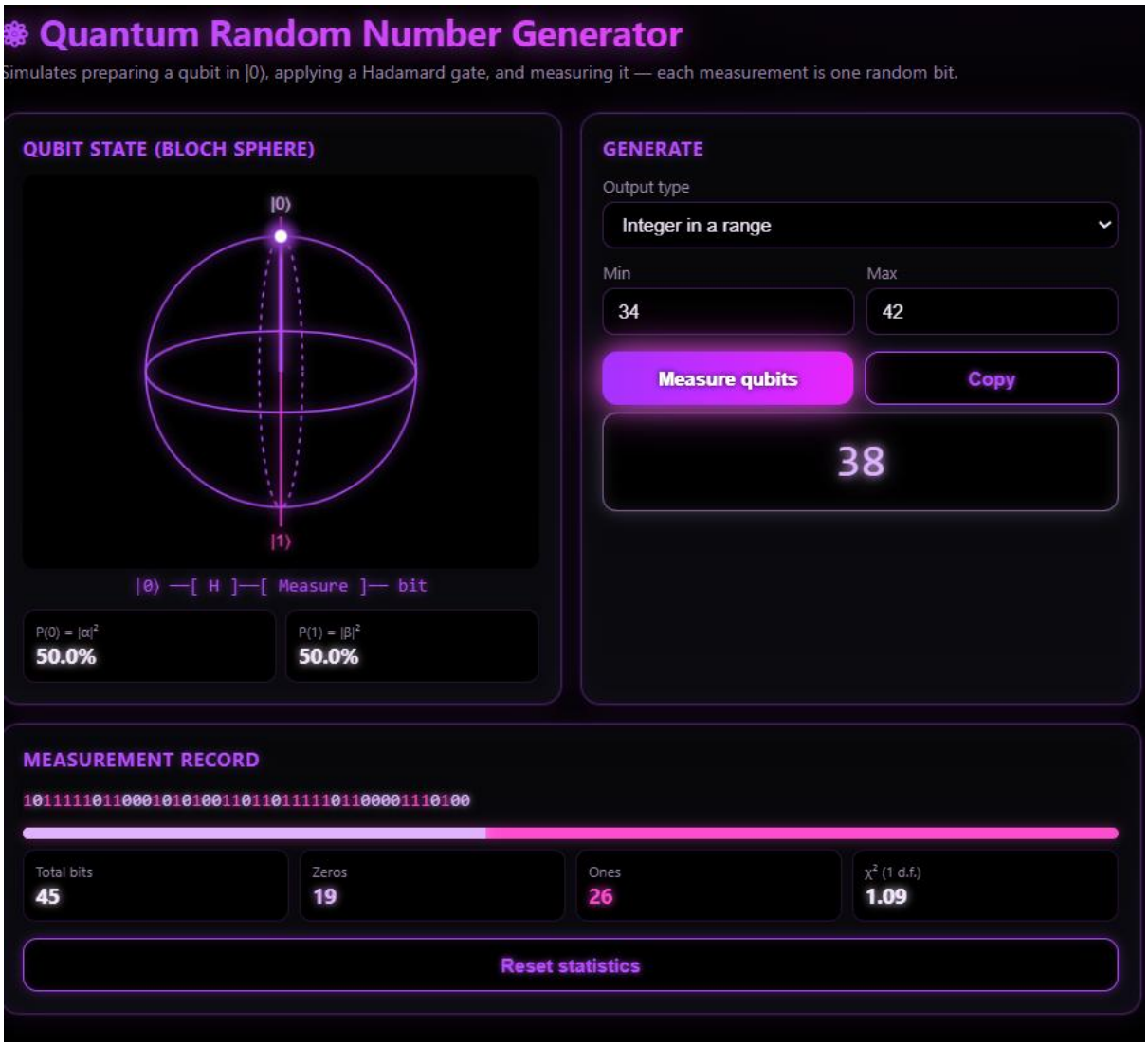
1.


Quantum Random Number Generator
Simulates preparing a qubit in |0⟩, applying a Hadamard gate, and measuring it — each measurement is one random bit.
QUBIT STATE (BLOCH SPHERE)
|0⟩
|1⟩
|0⟩ —[H]—[Measure]— bit
P(0) = |α|²
50.0%
P(1) = |β|²
50.0%
GENERATE
Output type
Integer in a range
Min
446
Max
1000
Measure qubits
Copy
675
MEASUREMENT RECORD
000000101110011010001111110011100101
Total bits
36
Zeros
18
Ones
18
χ² (1 d.f.)
0.00
Reset statistics
How it works: a qubit in state α|0⟩ + β|1⟩ is simulated as a 2-element complex state vector. A Hadamard gate takes |0⟩ to (|0⟩ + |1⟩)/√2, and a measurement collapses it to 0 or 1 with probability |α|² and |β|² (the Born rule).
Honest caveat: this runs on a classical computer, so the "collapse" is decided by the browser's cryptographically secure generator (crypto.getRandomValues). It is

2.


Quantum Random Number Generator
Simulates preparing a qubit in |0⟩, applying a Hadamard gate, and measuring it — each measurement is one random bit.
QUBIT STATE (BLOCH SPHERE)
|0⟩
|1⟩
|0⟩ —[H]—[Measure]— bit
P(0) = |α|²
50.0%
P(1) = |β|²
50.0%
GENERATE
Output type
Integer in a range
Min
44
Max
100
Measure qubits
Copy
95
MEASUREMENT RECORD
00000010111001101000111111001110010111001 1
Total bits
42
Zeros
20
Ones
22
χ² (1 d.f.)
0.10
Reset statistics
How it works: a qubit in state α|0⟩ + β|1⟩ is simulated as a 2-element complex state vector. A Hadamard gate takes |0⟩ to (|0⟩ + |1⟩)/√2, and a measurement collapses it to 0 or 1 with probability |α|² and |β|² (the Born rule).

3.


Quantum Random Number Generator
Simulates preparing a qubit in |0⟩, applying a Hadamard gate, and measuring it — each measurement is one random bit.
QUBIT STATE (BLOCH SPHERE)
|0⟩
|1⟩
|0⟩ —[H]—[Measure]— bit
P(0) = |α|²
50.0%
P(1) = |β|²
50.0%
GENERATE
Output type
Integer in a range
Min
34
Max
42
Measure qubits
Copy
38
MEASUREMENT RECORD
10111110110001010100110110111110110000111010 0
Total bits
45
Zeros
19
Ones
26
χ² (1 d.f.)
1.09
Reset statistics